

(TR-104) PYTHON WITH AI/ML

Training Day 92 Report

Date: 19 May 2026

Topic Covered: Voice Recognition Research, Speech Processing Libraries & Audio AI Techniques

The ninety-second day of training focused on researching voice recognition systems, exploring speech-processing libraries, and understanding techniques used in speech-to-text and audio analysis applications.

- Researched voice recognition technologies and their role in AI-powered applications.
- Explored speech-processing libraries including SpeechRecognition, Whisper and Librosa.
- Studied speech-to-text conversion workflows and audio preprocessing techniques.
- Analyzed noise reduction, feature extraction, and voice enhancement methods.
- Reviewed speaker recognition and voice authentication concepts.
- Explored real-time voice processing and transcription techniques.
- Studied the working principles of automatic speech recognition (ASR) systems.
- Compared different voice recognition models based on accuracy, speed, and deployment requirements.
- Analyzed practical applications of voice recognition in virtual assistants, accessibility solutions, and smart systems.
- Reviewed integration approaches for voice-based interaction within AI applications.
- Revised concepts related to speech recognition, audio processing, voice authentication, transcription systems, and AI-driven voice interfaces.

Overall, the training enhanced understanding of voice recognition technologies, speech-processing workflows, audio analysis techniques, and intelligent voice-enabled systems.